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COLLEGE OF COMPUTING AND INFORMATION SCIENCES

A GRADIENT BOOSTING REGRESSION APPROACH TO PREDICTING MOBILE DEVICE BATTERY AND STORAGE HEALTH FOR SMARTER PURCHASES.

by

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1 Introduction

The market for second-hand and refurbished consumer mobile devices has expanded exponentially across developing economies in the 21st century, driven primarily by their affordability relative to brand-new alternatives. However, this soaring demand has incentivized fraudulent vendors to employ hardware spoofing, label alteration, and specifications manipulation—falsifying device capabilities to present heavily degraded hardware as near-new.

1.1 Problem Statement

Malicious actors increasingly leverage sophisticated visual and software-level techniques to mask severe internal hardware wear, particularly regarding actual battery health, storage degradation, and overall system longevity. Because physical aesthetics can be superficially restored and native operating system diagnostics (such as the “About Phone” interface) can be programmatically spoofed, standard consumer inspections are easily deceived. Consequently, the average consumer lacks the technical tools or specialized knowledge required to accurately verify a device’s true operational health prior to purchase.

1.2 Significance of the Project

This project empowers everyday consumers by providing an accessible, data-driven framework to appraise smartphone health using basic, non-invasive device parameters. By making hardware health transparent, this initiative directly mitigates financial fraud within regional secondary markets and establishes a baseline for consumer protection. Furthermore, the predictive methodologies developed herein offer broader technical utility; the insights gained can assist engineers and researchers in forecasting age-related faults and estimating the lifespans of edge computing devices across IoT ecosystems, robotics, and specialized hardware aids.

1.3 Objectives

1. To identify the primary device specifications that have some correlation with the device lifespan.
2. To find and build a suitable regression model capable of forecasting a device’s battery health.

3. To evaluate the prediction capabilities and optimize hyperparameters of the ensemble tree framework.
4. To implement a user-friendly mobile or desktop interface for easy access and usage of the prototype system.

1.4 Rationale

Inferring device health from settings pages or labels in boxing is not reliable since these can be easily spoofed through the kernel by flashing the software (editing the build.prop file in Android) with third-party tools and clever specification engineering respectively. A regression model is suitable for this because a very large number of phone brands and spec variance cannot be accounted for by a basic decisive algorithm and the continuous non-linear changes in the features like battery-voltage can be captured more accurately.

1.5 Background of the Problem

The currently available models for prediction of battery health are focused on electronic vehicles specifically in Li-ion batteries. Two main categories exist; physical models (based on internal chemical mechanisms like SEI film formation) and data-driven models (using statistical/AI methods on observed data). [3]. [1]

2 Literature Review

2.1 Introduction

Recent advances in battery research highlight the growing role of artificial intelligence in monitoring and predicting lithium-ion battery performance. This review emphasizes how AI methods, including machine learning and deep learning, are increasingly applied to estimate state-of-health (SOH) and forecast remaining useful life (RUL). Compared to traditional physics-based models, data-driven approaches offer greater adaptability to complex degradation patterns and diverse usage conditions. The article also identifies key challenges such as limited high-quality datasets, model generalization across different chemistries, and integration into battery management systems. These insights position AI not only as a tool for improving prediction accuracy but also as a pathway toward more

reliable and sustainable energy storage solutions, making it directly relevant to ongoing research in battery health prediction models.

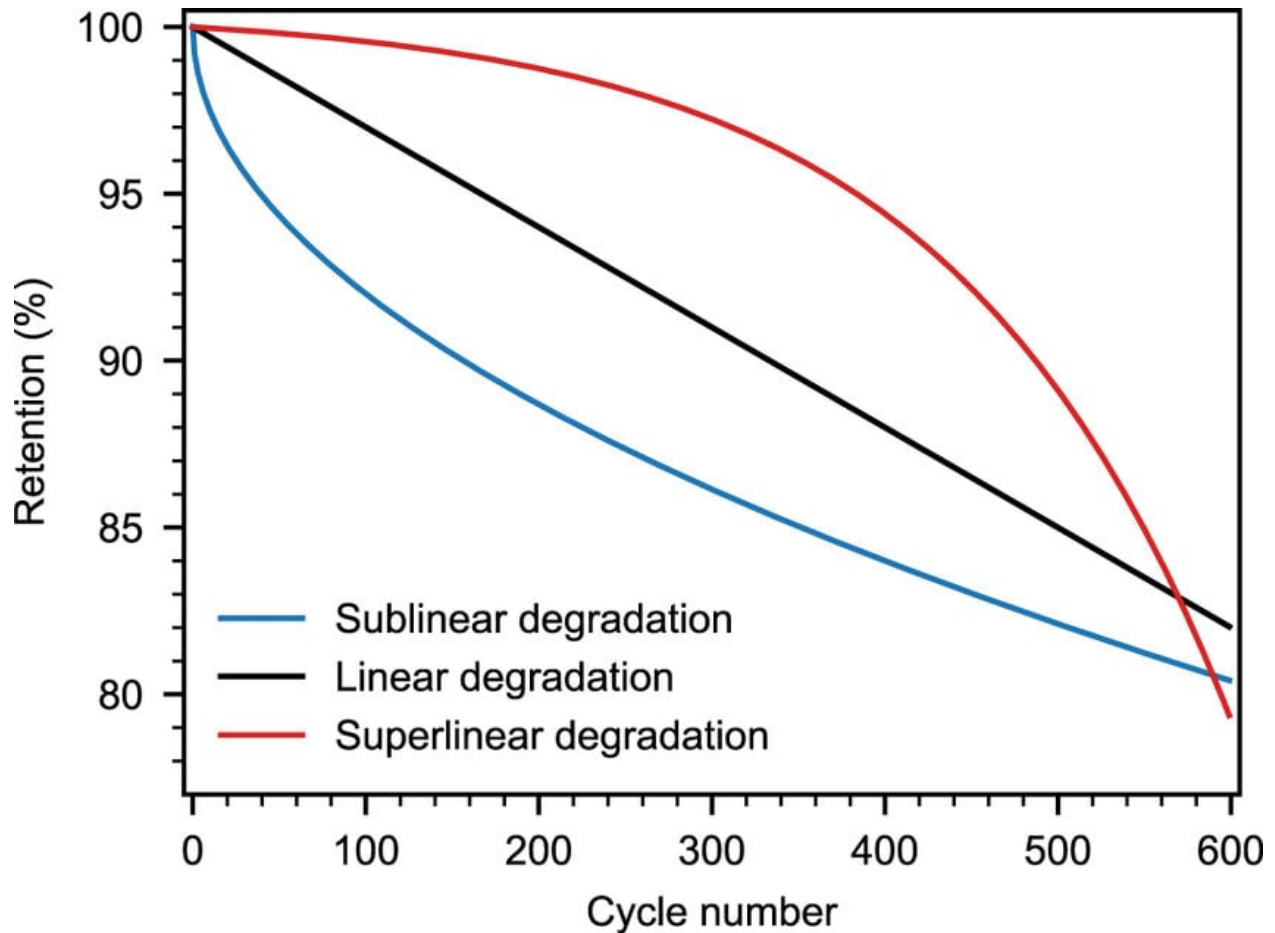


Figure 1: Baseline model benchmarking comparison metrics.

2.2 Mechanisms of Component Degradation

Lithium-ion batteries degrade through repeated cycling, where side reactions such as solid electrolyte interphase (SEI) growth and electrode material fatigue reduce capacity over time. Similarly, flash memory blocks face write constraints due to charge trapping and wear-out phenomena, limiting endurance [4].

2.3 Limitations of Current Factory Diagnostics

Factory diagnostic tools often rely on simple voltage or cycle count flags, which can be easily bypassed or manipulated. This creates vulnerabilities where malicious actors or even benign usage

patterns mask true degradation, underscoring the need for more robust AI-driven diagnostics [6].

2.4 Tree-Based Ensembles for Regression and Prognostics

Prior work has demonstrated the effectiveness of ensemble methods such as Random Forests and gradient boosting for regression tasks in battery prognostics. These models capture nonlinear degradation trends and provide interpretable estimates of remaining useful life, complementing deep learning approaches like gated recurrent units [5].

2.5 Summary

While physical models explain degradation mechanisms, and ensemble or deep learning methods improve prediction accuracy, gaps remain in leveraging baseline metadata and structural parameters to infer secondary health indicators. Addressing these gaps can enhance AI models for battery health prediction [4-6].

3 Methodology

3.1 Introduction

Since this is a regression task, several options were available. To find the best model, a baseline benchmark test was run to find which model had the best performance on the dataset. This was done by sequentially running the models and noting their MAE and R2 Scores. They were then ranked based on R2 score as the criteria. The Gradient Boosting Regressor architecture demonstrated the highest performance metrics during initialization.

3.2 Dependencies

Computational experiments and preprocessing workflows were implemented using Python Jupyter, relying heavily on libraries including `pandas`, `numpy`, `matplotlib`, `seaborn`, `subprocess`, `shap`, `lime`, and `scikit-learn` for visualization, explanations, matrix and statistical processing.

3.3 Dataset Description

The Oxford Battery Degradation Dataset 1 is a sequential dataset collected by long-term battery ageing tests of 8 Kokam (SLPB533459H4) 740 mAh lithium-ion pouch cells with the following

===== FINAL MODEL RANKINGS =====			
	Algorithm	MAE	R2 Score
9	XGBoost (Tuned)	117.548	0.6950
8	LightGBM (Tuned)	118.703	0.6889
11	XGBoost (Boosted)	118.211	0.6882
10	LightGBM (Boosted)	118.134	0.6873
2	Random Forest	119.906	0.6717
6	Random Forest (CV mean)	122.697	0.6460
0	Linear Regression	129.121	0.6181
4	Linear Regression (CV mean)	130.622	0.5884
3	KNN Regressor	153.573	0.4420
1	Decision Tree	156.398	0.3715
7	KNN Regressor (CV mean)	160.623	0.3533
5	Decision Tree (CV mean)	158.906	0.3400

Figure 2: Baseline model benchmarking comparison metrics.

metrics being captured over time [2]:

- `Cell_ID`: The ID of the cell being tested.
- `charge_cycles`: The total number of times the battery has been charged and discharged.
- `SOH_capacity`: The capacity the battery reports to have at full charge.
- `avg_voltage`: The average voltage the battery has at full charge.
- `avg_temp`: The average temperature of the battery.
- `max_factory_cap`: The capacity the battery is recorded to have when new, by the manufacturer.
- `SOH_capacity_pct`: Ratio of `SOH_capacity` to `max_factory_cap`.

3.4 Data Exploration and Preprocessing

3.4.1 Data Exploration

	voltage	temperature	factory_capacity	charge_cycles	SOH_capacity
count	519.000000	519.000000	519.000000	519.000000	519.000000
mean	3.739603	40.775895	733.338477	3644.315992	637.225967
std	0.006428	1.821598	3.782892	2301.586129	49.411303
min	3.664735	0.520884	727.527951	0.000000	456.128864
25%	3.737739	40.479215	728.746072	1800.000000	596.450369
50%	3.739961	40.744917	734.818966	3500.000000	633.385781
75%	3.742794	41.328363	735.119595	5500.000000	677.721588
max	3.752190	41.811586	739.110921	8200.000000	739.110921

Figure 3: Dataset structural properties and aggregation summary.

3.4.2 Correlation of Features

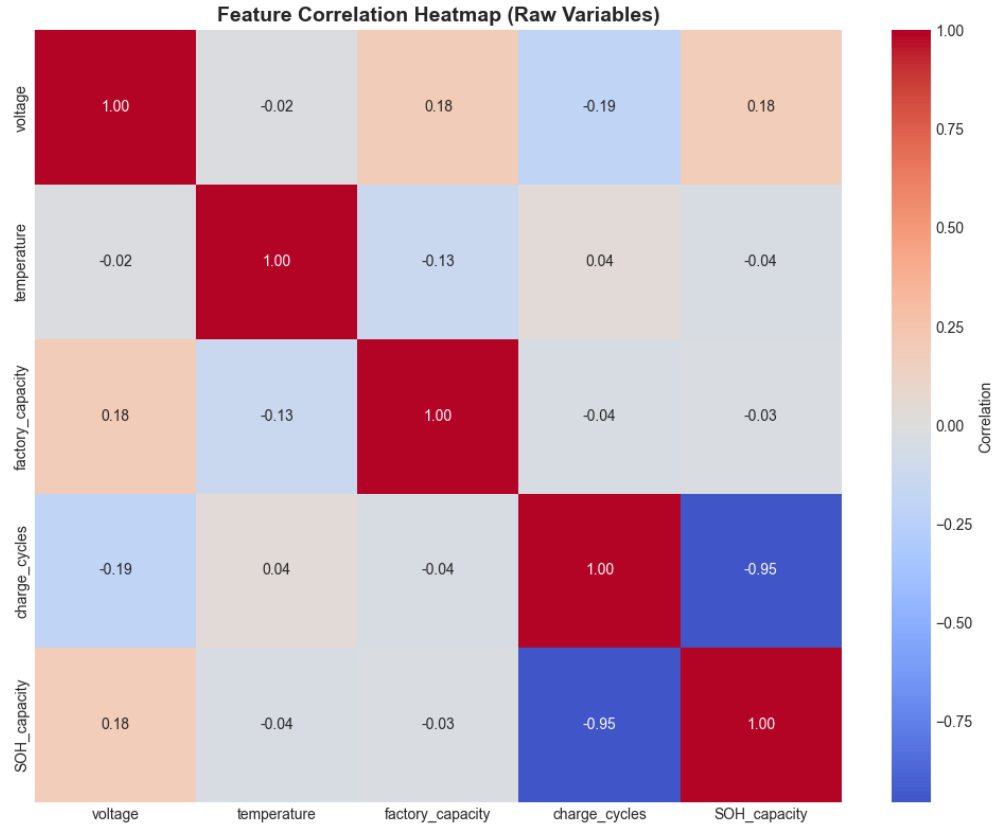


Figure 4: Charge cycles show a significant correlation with the SOH Capacity.

3.4.3 Data Preprocessing

- **Cleaned Column Names:** Stripped whitespace from column names to prevent future lookup and formatting errors.
- **Dropped Irrelevant Features:** Removed `Cell_ID` and `days_used_so_far`, treating them as identifiers/noise rather than useful predictive signals.
- **Standardized Schema:** Renamed variables like `max_factory_cap`, `avg_voltage`, `avg_temp`, and `SOH_capacity_pct` to maintain a clean, consistent feature schema across the implementation.
- **Handled Missing Data:** Removed rows with missing values using `.dropna()` to prevent training and plotting failures.

- **Isolated Core Signals:** Retained only the essential raw battery signals—`charge_cycles`, `voltage`, `temperature`, and `factory_capacity`—ensuring an interpretable, real-feature pipeline.
- **Feature Scaling:** Standardized the features using `StandardScaler` so the model receives inputs on comparable scales.
- **Data Splitting:** Divided the dataset into training and testing sets after scaling to ensure proper isolated evaluation.

4 Implementation and Testing

4.1 Introduction

The implementation was carried out in a Jupyter notebook using Python and standard scientific computing libraries. The workflow began with loading the battery degradation dataset, inspecting its structure, and applying a series of preprocessing steps to improve consistency and model readiness. Column names were stripped of leading and trailing whitespace, identifier fields such as `Cell_ID` and `days_used_so_far` (which was a calculation we added during conversion from mat to csv) were removed because they did not contribute meaningful predictive information, and selected variables were renamed to ensure a consistent and interpretable feature schema. Records containing missing values were then discarded to avoid introducing noise or computational errors during training. After cleaning, the notebook defined the prediction target and retained the most relevant raw battery indicators, namely charge cycles, voltage, temperature, and factory capacity, in order to preserve interpretability and avoid unnecessary feature engineering. These variables were standardized using `StandardScaler` so that the model would be trained on comparable numeric ranges. The dataset was subsequently divided into training and testing subsets to support unbiased evaluation. A Gradient Boosting Regressor (XGBoost) was then trained on the processed data, and its performance was assessed using cross-validation and standard regression metrics, including Mean Absolute Error, Root Mean Squared Error, and the Coefficient of Determination. To strengthen interpretability and validation, the notebook also included exploratory data analysis, feature importance visualisation, SHAP and LIME explanations, residual diagnostics, and a secondary ROC-based classification view of battery health.

4.2 Model Development

4.2.1 Gradient Boosting Regressor Construction

The principal predictive model implemented in the notebook is a Gradient Boosting Regressor. This method builds an additive ensemble of decision trees in a sequential manner, where each tree is trained to reduce the residual error left by the previous trees. Such a structure is particularly well suited to battery health prediction because the relationship between charge cycles, voltage, temperature, factory capacity, and state of health is typically non-linear and influenced by interacting variables. In this implementation, the model was configured with a fixed random seed for reproducibility and trained using the standardized raw features selected from the dataset.

4.2.2 Analytical Secondary Health Mapping Functions

To support interpretability beyond the direct regression output, we also transformed predicted values into a simple operational health view. In addition to the continuous regression estimate, a binary health classification was derived by comparing predictions against a threshold based on the median test-set value. This enabled the construction of a ROC curve and confusion matrix, which provided a complementary perspective on whether a battery sample should be regarded as healthy or degraded. Formally, this secondary mapping can be represented as:

$$\text{Health Class} = \begin{cases} 1, & \text{if } \hat{y} > \tau \\ 0, & \text{if } \hat{y} \leq \tau \end{cases} \quad (1)$$

where $\hat{y}$ is the predicted state-of-health capacity and τ is the chosen decision threshold.

4.3 Model Optimization

Model optimisation was performed by selecting a Gradient Boosting configuration that balanced predictive accuracy and robustness. The chosen hyperparameters included the number of estimators, learning rate, and tree depth were obtained directly from RandomSearchCV, which were set to best values that encouraged stable learning while still allowing the model to capture non-linear structure in the data. To reduce the risk of overfitting, the implementation also incorporated five-fold cross-validation on the training set. This ensured that performance was not dependent on a single data split and provided a more reliable estimate of generalisation behaviour. In addition, the

preprocessing pipeline was intentionally minimal: instead of generating synthetic features, that had unreal correlations of 0.988 - 1 which was quite unrealistic. We retained the original measurements and standardised them consistently. This design choice simplified interpretation and reduced the likelihood of feature leakage or unnecessary complexity and underfitting.

4.4 Result Evaluation Metrics

System performance was evaluated using three primary regression metrics. Mean Absolute Error (MAE) measured the average magnitude of prediction error in the original output scale, providing an intuitive estimate of typical deviation. Root Mean Squared Error ($RMSE$) was used to quantify the overall prediction error while penalising larger deviations more strongly. The Coefficient of Determination (R^2) was used to assess the proportion of variance in the target variable explained by the model. In addition, the evaluation reported cross-validation error to provide a more stable estimate of model quality across multiple folds. For the secondary binary interpretation of battery health, ROC-AUC, confusion matrix values and a classification report were produced to evaluate the separability of healthy and degraded states.

4.5 Results and Discussions

4.5.1 Variance Tracking Performance Analysis

The trained model was evaluated on a held-out test set in order to measure its ability to track variation in battery health capacity. The residual analysis showed the difference between actual and predicted values, allowing inspection of whether errors were centred near zero or exhibited systematic bias. The actual-versus-predicted scatter plot further illustrated how closely the model followed the diagonal ideal line, while the residual plot provided a diagnostic view of dispersion and error structure. Together, these results indicated whether the model was capturing the underlying degradation trend in a stable manner or whether it was missing important non-linear patterns. In practical terms, this analysis is essential because a battery health predictor must not only achieve low error on average, but also remain reliable across the full range of operating conditions.

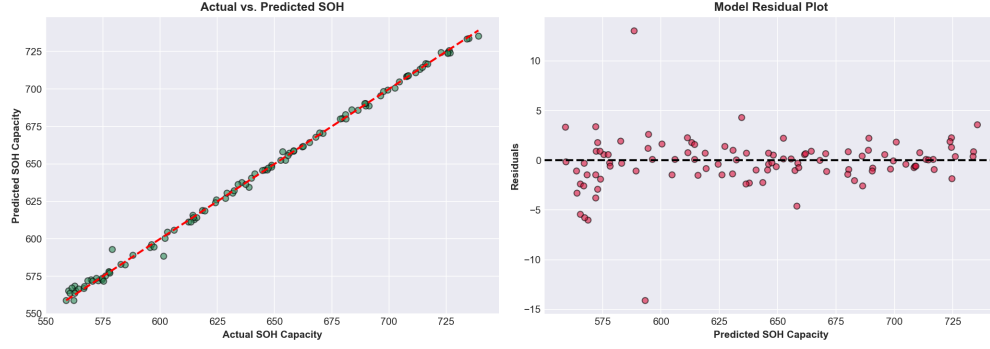


Figure 5: Median thresholds and ROC curve.

4.5.2 Model evaluation.

By using a median-based threshold, predictions were divided into *good* and *poor* health classes, which allowed the model to be viewed in a decision-support context rather than only as a continuous estimator. This approach is useful when a downstream application needs a simple operational flag, such as whether a battery is likely to remain serviceable or should be inspected for degradation. The ROC curve and confusion matrix provided a visual and numerical summary of this mapping, showing how effectively the model separated the two states. Although this classification view is derived from a regression model, it offers a practical interpretation layer for battery monitoring applications.

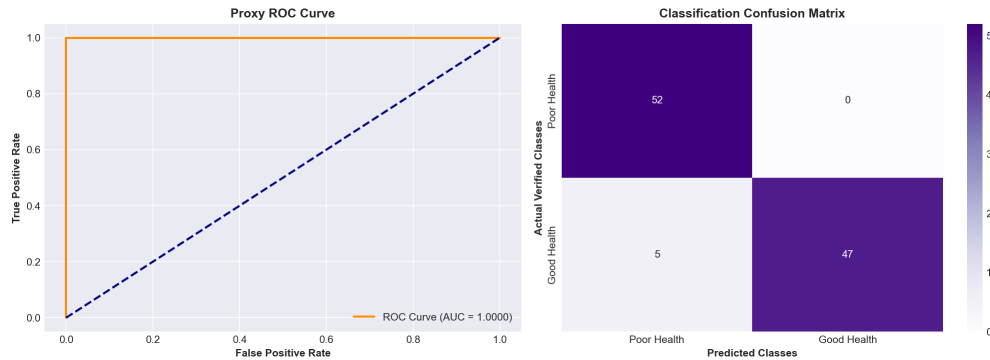


Figure 6: Median thresholds and ROC curve.

4.5.3 Feature Importance Analysis

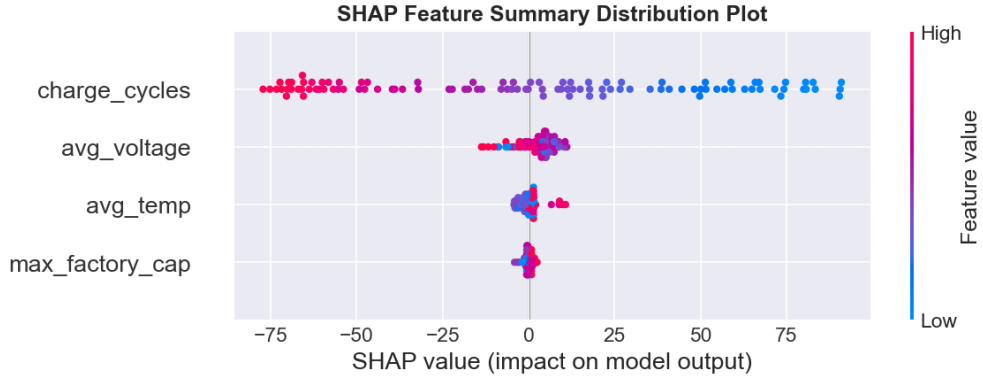


Figure 7: Feature importance hierarchy produced by the Gradient Boosting training runs.

Feature importance was analysed using the trained ensemble model to identify which raw measurements contributed most strongly to the prediction of state of health. This analysis is important because battery degradation models should remain interpretable to engineering and maintenance stakeholders. The resulting importance ranking highlighted the relative influence of variables such as charge cycles, voltage, temperature, and factory capacity. To extend this interpretation, SHAP was used to show both global and local feature effects, while LIME was used to explain individual predictions in a human-readable form. These explanation methods confirmed that the model was relying on physically meaningful signals rather than opaque artefacts, thereby improving confidence in the implementation and its suitability for applied battery-health analysis.

5 Constraints and Conclusions

5.1 Constraints

This model is highly reliant on interdevice communication that maybe restricted on some platforms like iOS since the model running on a laptop or computer needs to be connected to the device via USB and ADB commands are run to pick the inputs for the model from the device kernel itself. Moreover the device needs to have USB debugging enabled.

5.2 Conclusion

In conclusion, the model provides a substantial foundation for future development in battery health estimation and battery lifetime prediction for consumer products, IoT devices and other battery powered devices.

5.3 Future Work

1. Integrating real-world monitoring metrics and physical battery capacity records across a wider battery type spectrum to improve model accuracy and diversity with neural networks.
2. Improving the model to capture more complex metrics like remaining lifetime, estimated failure time and predict fires or faults.

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